

Task 01 – Differential Motion Analysis

For this task, we implemented motion detection using simple frame subtraction:

$$\text{Diff} = |I_{t+1} - I_t|$$

Both frames were converted to grayscale and smoothed using Gaussian blur. We tested the method using image pairs that include noise, small motion, large motion, and illumination changes.

Instead of using a fixed threshold, we used an adaptive threshold based on Median Absolute Deviation (MAD):

$$T = k \times \sigma$$

where ($\sigma \approx 1.4826 \times \text{MAD}$). We selected $k = 4$ because smaller values detected noise as motion, while larger values missed small movements. In most cases, the threshold remained between 10 and 80, which produced stable results.

The method worked well when the camera was stable and lighting changes were small. However, it was less successful under strong illumination variation or camera shake.

This approach only detects motion regions and does not provide direction. To include motion direction, we can use optical flow or block matching to estimate motion vectors $((u, v))$.

Task 02 – Fast Optical Flow (Lucas–Kanade)

In this task, we implemented dense optical flow using the Lucas–Kanade method. Unlike frame subtraction, optical flow estimates the motion vector $((u, v))$ for each pixel, which gives both the magnitude and direction of motion.

The method is based on the brightness constancy assumption:

$$I_x u + I_y v + I_t = 0$$

We computed spatial gradients using the Scharr operator and the temporal gradient from frame differences. To make the computation faster, we used box filtering to efficiently calculate window sums instead of looping through each pixel window.

To avoid unstable regions (aperture problem), we ignored pixels where the determinant of the structure tensor was very small.

We visualized the flow using HSV color coding, where color represents direction and brightness represents motion magnitude.

The method worked well for small and smooth motion. However, it struggled with large motion between frames and in low-texture regions. Compared to simple frame subtraction, optical flow provides much more information because it includes motion direction as well as magnitude.